

If treatment is discontinued for more than 2 days, reinstitute with 12.5 mg once or twice daily; if that dose is tolerated, the dose may be increased to the previously therapeutic dose more quickly than recommended for initial treatment

If abrupt discontinuation of clozapine is necessary, the patient must be covered for cholinergic rebound; those with higher clozapine plasma levels may need extremely high doses of anticholinergic medications to prevent delirium and other rebound symptoms

Slow off-titration is preferred if possible to avoid cholinergic rebound and rebound psychosis

Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly

Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine

For treatment resistance, consider obtaining plasma drug levels to guide dosing above recommended levels

Overdose

- Sometimes lethal; changes in heart rhythm, excess salivation, respiratory depression, altered state of consciousness

Long-Term Use

- Treatment to reduce risk of suicidal behavior should be continued for at least 2 years
- Medication of choice for treatment-refractory schizophrenia

Habit Forming

- No

How to Stop

- See The Art of Switching section of individual agents for how to stop clozapine, generally over at least 4 weeks
- See Tables for guidance on stopping due to neutropenia
- * Rapid discontinuation may lead to rebound psychosis and worsening of symptoms

Pharmacokinetics

- Half-life 5–16 hours
- Metabolized primarily by CYP450 1A2 and to a lesser extent by CYP450 2D6 and 3A4



Drug Interactions

- Use clozapine plasma levels to guide treatment due to propensity for drug interactions

- In the presence of a strong CYP450 1A2 inhibitor (e.g., fluvoxamine, ciprofloxacin): use 1/3 the dose of clozapine
- In the presence of a strong CYP450 1A2 inducer (e.g., cigarette smoke), clozapine plasma levels are decreased
- May need to decrease clozapine dose by up to 50% during periods of extended smoking cessation (>1 week)
- Strong CYP450 2D6 inhibitors (e.g., bupropion, duloxetine, paroxetine, fluoxetine) can raise clozapine levels; dose adjustment may be necessary
- Strong CYP450 3A4 inhibitors (e.g., ketoconazole) can raise clozapine levels; dose adjustment may be necessary
- Clozapine may enhance effects of antihypertensive drugs
- May antagonize levodopa, dopamine agonists

Other Warnings/Precautions

- Use with caution in patients on other anticholinergic agents (benztropine, trihexyphenidyl, olanzapine, quetiapine, chlorpromazine, oxybutynin, and other antimuscarinics)
- Constipation caused by clozapine can uncommonly lead to serious bowel complications, so advise patients of the risk and the need to stay hydrated, question patients about their bowel movements throughout treatment, and advise them to contact a healthcare professional right away if they have difficulty having a bowel

movement, do not have a bowel movement at least three times a week or less than their normal frequency, or are unable to pass gas

- Should not be used in conjunction with agents that are known to cause neutropenia
- Myocarditis is rare and only occurs in the first 6 weeks of treatment
- Cardiomyopathy is a late complication (consider annual ECG)
- Use with caution in patients with glaucoma
- Use with caution in patients with enlarged prostate
- Use with caution in patients with conditions that predispose to hypotension (dehydration, overheating)
- Atypical antipsychotics have the potential for cognitive and motor impairment, especially related to sedation
- Atypical antipsychotics can cause body temperature dysregulation; use caution in patients who may experience conditions that increase body temperature (e.g., strenuous exercise, saunas, extreme heat, dehydration, or concomitant anticholinergic medication)
- Dysphagia has been associated with antipsychotic use, and clozapine should be used cautiously in patients at risk for aspiration pneumonia
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic

treatment, and recurrently for patients on long-term antipsychotic therapy

- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- In patients with myeloproliferative disorder
- In patients with uncontrolled epilepsy
- In patients with paralytic ileus
- In patients with CNS depression
- If there is a proven allergy to clozapine

Special populations

Renal Impairment

- Should be used with caution

Hepatic Impairment

- Should be used with caution

Cardiac Impairment

- Use in patients with cardiac impairment has not been studied, so use with caution because of risk of orthostatic hypotension

- Use with caution if patient is taking concomitant antihypertensive or alpha 1 antagonist

Elderly

- Some patients may tolerate lower doses better
- Although atypical antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with atypical antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events
- Consider pimavanserin for dementia-related psychosis or Parkinson's disease psychosis instead of an atypical antipsychotic



Children and Adolescents

- Safety and efficacy have not been established
- Preliminary research has suggested efficacy in early-onset treatment-resistant schizophrenia
- Children and adolescents using clozapine may need to be monitored more often than adults and may tolerate lower doses better



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Animal studies have not shown adverse effects
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Clozapine should be used only when the potential benefits outweigh potential risks to the fetus
- National Pregnancy Registry for Atypical Antipsychotics: 1-866-961-2388,
<https://womensmentalhealth.org/research/pregnancyregistry/atypicalantipsychotic>

Breast Feeding

- Unknown if clozapine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed
- Infants of women who choose to breast feed while on clozapine should be monitored for possible adverse effects

The art of psychopharmacology

Potential Advantages

- * Treatment-resistant schizophrenia
- * Violent, aggressive patients
- * Patients with tardive dyskinesia
- * Patients with suicidal behavior

Potential Disadvantages

- * Patients with diabetes, obesity, and/or dyslipidemia
- Sialorrhea, sedation, and orthostasis may be intolerable for some

Primary Target Symptoms

- Positive symptoms of psychosis
- Negative symptoms of psychosis
- Cognitive symptoms

- Unstable mood (both depression and mania)
- Suicidal behavior
- Aggressive symptoms

Pearls

- * Clozapine is the gold standard treatment for refractory schizophrenia
- * Clozapine is not used first-line due to side effects and monitoring burden
- * However, some studies have shown that clozapine was associated with the lowest risk of mortality among the antipsychotics, causing the study authors to question if its use should continue to be restricted to resistant cases
- May reduce violence and aggression in difficult cases, including forensic cases
- * Reduces suicide in schizophrenia
- May reduce substance abuse
- May improve tardive dyskinesia
- Little or no prolactin elevation, motor side effects, or tardive dyskinesia
- Clinical improvements often continue slowly over several years
- Cigarette smoke can decrease clozapine levels and patients may be at risk for relapse if they begin or increase smoking

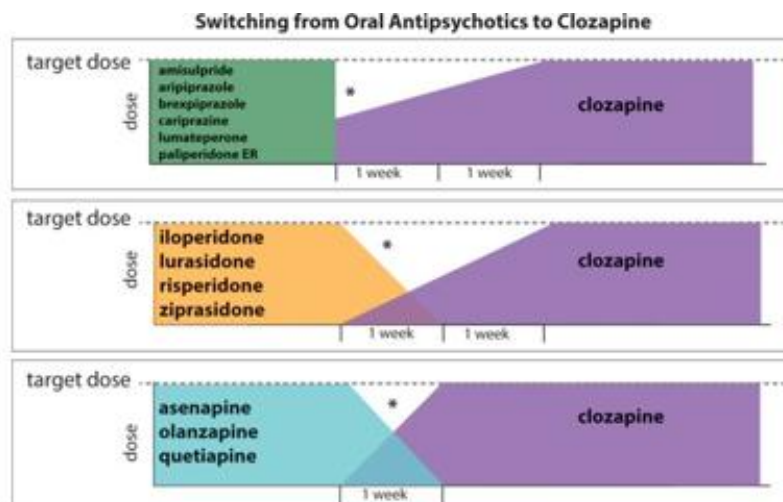
- More weight gain than many other antipsychotics –does not mean every patient gains weight
- Patients can have much better responses to clozapine than to any other agent, but not always
- For treatment-resistant patients, especially those with impulsivity, aggression, violence, and self-harm, long-term polypharmacy with 2 atypical antipsychotics or with 1 atypical antipsychotic and 1 conventional antipsychotic may be useful or even necessary while closely monitoring
- In such cases, it may be beneficial to combine 1 depot antipsychotic with 1 oral antipsychotic
- To treat constipation and reduce risk of paralytic ileus and bowel obstruction, taper off other anticholinergic agents and start all patients routinely on docusate
- The US FDA has changed the requirements for monitoring, prescribing, dispensing, and receiving clozapine in order to address concerns related to neutropenia; in addition to updating the prescribing information for clozapine, the FDA has approved a new, shared Risk Evaluation and Mitigation Strategy (REMS)
- The Clozapine REMS program replaces the six existing clozapine registries, which are maintained by individual clozapine manufacturers. Prescribers, pharmacies, and patients will now be required to enroll in a single centralized program; patients already treated with clozapine will be automatically transferred. In order to prescribe and dispense clozapine, prescribers and pharmacies will be

required to be certified in the Clozapine REMS program. Visit the Clozapine REMS program homepage for more information.

The Art of Switching

Switching from Oral Antipsychotics to Clozapine

- With amisulpride, aripiprazole, brexpiprazole, cariprazine, lumateperone, and paliperidone ER, immediate stop is possible; begin clozapine at middle dose
 - With risperidone, ziprasidone, iloperidone, and lurasidone, begin clozapine gradually, titrating over at least 2 weeks to allow patients to become tolerant to the sedating effect
- * Benzodiazepine or anticholinergic medication can be administered during cross-titration to help alleviate side effects such as insomnia, agitation, and/or psychosis. However, use with caution in combination with clozapine as this can increase the risk of circulatory collapse



Suggested Reading

Huhn M , Nikolakopoulou A , Schneider-Thoma J , et al. Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of adults with multi-episode schizophrenia: a systematic review and network meta-analysis . *Lancet* 2019 ;**394** :939 -51 .

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Masuda T , Misawa F , Takase M , Kane JM , Correll CU . Association with hospitalization and all-cause discontinuation among patients with schizophrenia on clozapine vs other oral second-generation antipsychotics: a systematic review and meta-analysis of cohort studies . *JAMA Psychiatry* 2019 ;**76** :1052 –62 . doi: 10.1001/jamapsychiatry.2019.1702 .

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Ronaldson KJ , Fitzgerald PD , McNeil JJ . Clozapine-induced myocarditis, a widely overlooked adverse reaction . *Acta Psychiatr Scand* 2015 ;**132** :231–40 .

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Rosenheck RA , Davis S , Covell N , et al. Does switching to a new antipsychotic improve outcomes? Data from the CATIE Trial . *Schizophr Res* 2009 ;**170** (1):22 –9.

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Schulte P . What is an adequate trial with clozapine?: therapeutic drug monitoring and time to response in treatment-refractory schizophrenia . *Clin Pharmacokinet* 2003 ;**42** :607–18 .

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Tiihonen J , Lonnqvist J , Wahlbeck K , et al. 11-year follow-up of mortality in patients with schizophrenia: a population-based cohort study (FIN11

study) . *Lancet* 2009 ; **374** (9690):620 –7 .

Cyamemazine

Therapeutics

Brands

- Tercian
- *see index for additional brand names*

Generic?

Not in USA

Class

- Conventional antipsychotic (neuroleptic, phenothiazine, dopamine 2 antagonist, serotonin dopamine antagonist)

Commonly Prescribed for

(bold for FDA approved)

- Schizophrenia
- * Anxiety associated with psychosis (short-term)
- Anxiety associated with nonpsychotic disorders, including mood disorders and personality disorders (short-term)
- Severe depression
- Bipolar disorder

- Other psychotic disorders
- Acute agitation/aggression (injection)
- Benzodiazepine withdrawal

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis
- * Although classified as a conventional antipsychotic, cyamemazine is a potent serotonin 2A antagonist
- Affinity at a myriad of other neurotransmitter receptors may contribute to cyamemazine's efficacy
- * Specifically, antagonist actions at 5HT2C receptors may contribute to notable anxiolytic effects in many patients
- 5HT2C antagonist actions may also contribute to antidepressant actions in severe depression and to improvement of cognitive and negative symptoms of schizophrenia in some patients

How Long Until It Works

- Psychotic symptoms can improve with high doses within 1 week, but it may take several weeks for full effect on behavior
- Anxiolytic actions can improve with low doses within 1 week, but it may take several days to weeks for full effect on behavior

If It Works

- High doses most often reduce positive symptoms in schizophrenia but do not eliminate them
- Low doses most often reduce anxiety symptoms in psychotic and nonpsychotic disorders
- Most schizophrenia patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Continue treatment in schizophrenia until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year, after first episode of psychosis in schizophrenia
- For second and subsequent episodes of psychosis in schizophrenia, treatment may need to be indefinite
- For symptomatic treatment of anxiety in psychotic and nonpsychotic disorders, treatment may also need to be indefinite while monitoring the risks versus the benefits of long-term treatment
- Reduces symptoms of acute psychotic mania but not proven as a mood stabilizer or as an effective maintenance treatment in bipolar disorder
- After reducing acute psychotic symptoms in mania, consider switching to a mood stabilizer and/or an atypical antipsychotic for long-term mood stabilization and maintenance

If It Doesn't Work

- For treatment of psychotic symptoms, consider trying one of the first-line atypical antipsychotics (risperidone, olanzapine, quetiapine, ziprasidone, aripiprazole, paliperidone, amisulpiride)
- Consider trying another conventional antipsychotic
- If 2 or more antipsychotic monotherapies do not work, consider clozapine
- For treatment of anxiety symptoms, consider adding a benzodiazepine or switching to a benzodiazepine

Best Augmenting Combos for Partial Response or Treatment Resistance

- Generally, best to switch to another agent
- Augmentation of conventional antipsychotics has not been systematically studied
- Addition of a mood-stabilizing anticonvulsant such as valproate, carbamazepine, or lamotrigine may be helpful in both schizophrenia and bipolar mania
- Augmentation with lithium in bipolar mania may be helpful
- Addition of a benzodiazepine, especially for short-term agitation
- Addition of antidepressants for severe depression

Tests

* Since conventional antipsychotics are frequently associated with weight gain, before starting treatment, weigh all patients and determine if the patient is already overweight (BMI 25.0–29.9) or obese BMI >30)

- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting glucose 100–125 mg/dL), diabetes (fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

* Monitor weight and BMI during treatment

* Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics

* While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or dyslipidemia, or consider switching to a different antipsychotic

- Should check blood pressure in the elderly before starting and for the first few weeks of treatment
- Monitoring elevated prolactin levels of dubious clinical benefit
- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count

(CBC) monitored frequently during the first few months and cyamemazine should be discontinued at the first sign of decline of WBC in the absence of other causative factors

Side effects

How Drug Causes Side Effects

- By blocking dopamine 2 receptors in the striatum, it can cause motor side effects at antipsychotic (high) doses
- Much lower propensity to cause motor side effects at low doses used to treat anxiety
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin, but unlike other conventional antipsychotics, prolactin elevations at low doses of cyamemazine are uncommon or transient
- By blocking dopamine 2 receptors excessively in the mesocortical and mesolimbic dopamine pathways, especially at high doses, it can cause worsening of negative and cognitive symptoms (neuroleptic-induced deficit syndrome)
- Anticholinergic actions, especially at high doses, may cause sedation, blurred vision, constipation, dry mouth
- Antihistamine actions may contribute to anxiolytic actions at low doses and to sedation and weight gain at high doses

- By blocking alpha 1 adrenergic receptors, cyamemazine can cause dizziness, sedation, and hypotension especially at high doses
- Mechanism of weight gain and any possible increased incidence of diabetes and dyslipidemia with conventional antipsychotics is unknown

Notable Side Effects

- * Neuroleptic-induced deficit syndrome (unusual at low doses)
- Akathisia
- Drug-induced parkinsonism, tardive dyskinesia (unusual at low doses)
- Galactorrhea, amenorrhea (unusual at low doses)
- Hypotension, tachycardia (unusual at low doses)
- Dry mouth, constipation, vision disturbance, urinary retention
- Sedation
- Decreased sweating
- Weight gain (may be unusual at low doses)
- Sexual dysfunction
- Metabolic effects, glucose tolerance

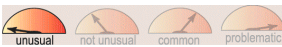


Life-Threatening or Dangerous Side Effects

- Rare neuroleptic malignant syndrome

- Rare seizures
- Rare jaundice, agranulocytosis
- Increased risk of death and cerebrovascular events in elderly patients with dementia-related psychosis

Weight Gain



- Reported but not expected especially at low doses

Sedation



- Many experience and/or can be significant in amount, especially at high doses
- Sedation is usually dose-dependent and may not be experienced as sedation but as anxiolytic actions on anxiety and aggression at low doses where cyamemazine may function as an atypical antipsychotic (e.g., <300 mg/day; especially 25–100 mg/day)

What to Do About Side Effects

- Wait
- Wait
- Wait
- For motor symptoms, add an anticholinergic agent

- Reduce the dose
- For sedation, give at night
- Switch to an atypical antipsychotic
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia

Best Augmenting Agents for Side Effects

- Benztropine or trihexyphenidyl for motor side effects
- Benzodiazepines may be helpful for akathisia
- Many side effects cannot be improved with an augmenting agent

Dosing and use

Usual Dosage Range

- 50–300 mg at bedtime for treatment of psychosis
- 25–100 mg for anxiety; duration of treatment 4 weeks
- Children (ages 6 and older): 1–4 mg/kg per day
- Injection: 25–100 mg/day

Dosage Forms

- Tablet 25 mg, 100 mg
- Oral solution 40 mg/mL

- Injection 50 mg/5 mL

How to Dose

- Psychosis: usual maintenance dose 50–300 mg at bedtime; maximum dose 600 mg/day divided into 2 or 3 doses; after 2 weeks consider reducing to lowest effective dose
- Anxiety (adults): usual dose 25–100 mg/day; reduce dose if unacceptable sedation; maximum duration of treatment 4 weeks
- Anxiety (children): usual dose 1–4 mg/kg per day



Dosing Tips

- Has conventional antipsychotic properties at originally recommended high doses (300–600 mg/day)
- * Binding studies, PET studies, and clinical observations suggest that cyamemazine may be “atypical” with low motor side effects or prolactin elevations at low doses (below 300 mg/day)
- * Clinical evidence suggests substantial anxiolytic benefits at 25–100 mg/day in many patients
- * Clinical evidence suggests low risk of drug-induced parkinsonism, little prolactin elevation yet demonstrable anxiolytic, anti-aggression, and antidepressant actions at doses below 300 mg/day
- Robust antipsychotic actions on positive symptoms may require dosing above 300 mg/day

- Low doses up to 100 mg/day may be used to augment partial responders to other conventional or atypical antipsychotics, especially for anxiolytic actions
- Treatment should be suspended if absolute neutrophil count falls below 1000/mm³

Overdose

- Drug-induced parkinsonism, sedation, hypotension, coma, respiratory depression

Long-Term Use

- Some side effects may be irreversible (e.g., tardive dyskinesia)

Habit Forming

- No

How to Stop

- Slow down-titration (over 6–8 weeks), especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)
- Rapid oral discontinuation of high doses of phenothiazines in psychotic patients may lead to rebound psychosis and worsening of symptoms

- If antiparkinsonian agents are being used, they should generally be continued for a few weeks after high-dose cyamemazine is discontinued

Pharmacokinetics

- Half-life 10 hours



Drug Interactions

- May decrease the effects of levodopa; contraindicated for use with dopamine agonists other than levodopa
- May increase the effects of antihypertensive drugs except for guanethidine, whose antihypertensive actions phenothiazines may antagonize
- May enhance QTc prolongation of other drugs capable of prolonging QTc interval
- Additive effects may occur if used with CNS depressants
- Anticholinergic effects may occur if used with atropine or related compounds
- Some patients taking a neuroleptic and lithium have developed an encephalopathic syndrome similar to neuroleptic malignant syndrome
- Epinephrine may lower blood pressure; diuretics and alcohol may increase risk of hypotension when administered with a phenothiazine